

Robust Battery Management System for Electric Vehicles

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Abstract—Battery Management System (BMS) is an intrinsic module in electric vehicles (EVs), which ensures safe, efficient and economical operation by monitoring the real time battery status and safeguards the system from overcharging and undercharging. The state of charge (SOC) and state of health (SOH) are the two key decision making, battery health monitoring parameters in a BMS. This paper addresses the problem of estimation of state of charge (SOC) and state of health (SOH) of batteries for BMS used in EVs. The SOC and SOH are estimated using Unscented Kalman filter (UKF) algorithm. A novel adaptive algorithm, derived based on Lyapunov theory, has been proposed to tune the threshold on SOC for diverse operating conditions. The algorithm has been designed in such a way that the knowledge of SOH is not required for tuning the threshold on SOC adaptively.

Index Terms—BMS, State of Charge, State of Health, UKF, Adaptive threshold

I. INTRODUCTION

ELECTRIC vehicles (EVs) have revolutionized the transportation sector during the past one decade. A robust BMS to guarantee the optimal performance of cells is inevitable for the safety and reliability of the Evs. Taking into account the factors like load on battery, operating cell voltages, operational current and temperature data captured through sensors and even estimated parameters like SOC and SOH, the BMS determines charging/discharging cycles of the battery cells. Numerous battery management architectures are depicted in literature.

In [1], Xiong and Shen have proposed extended Kalman filter (EKF)-based short circuit (SC) fault detection method in EV batteries. Here, the SOC of the battery cells are estimated using Extended Kalman Filter(EKF) on the basis of real-time data collected using sensors and the SC resistance value is computed based on the calculated and estimated SOC values.

In [2], Shen *et.al* presented a co-estimation strategy for SOC based on EKF and that for SOH, and state of function (SOF) based on the recursive Least Square (RLS) estimation for a real time BMS. The battery capacity and the power output are then computed online based on these estimates, where a predefined constant thresholds for these parameters are used to identify the faults. In [3], authors proposed a robust novel estimation scheme for SOH based on the real

time constant current charging and discharging profiles of the batteries, which is robust to diverse battery operating conditions, but the parameters such as SOC and SOF are not considered. An extreme learning machine neural architecture is proposed in [4] for fault detection in EV batteries using the voltage signals, where Genetic algorithm-based method is used to tune the weights and the thresholds on weights. Similar work has been done in [5], where an LSTM-based neural architecture and a equivalent circuit of the battery is used for fault identification, which claims robust fault assessment and can give fore-warning of thermal runaways, but a detailed analysis of the battery status is not guaranteed in this work.

In [6], unscented Kalman filter (UKF) algorithm has been used to compute the SOC and to forecast the maximum power output and the power distribution. Similar SOC estimation strategies based on EKF and UKF are adopted in [8] and [9]. An adaptive UKF algorithm with increased estimation accuracy and stability is proposed in [10].

In [11], Yu presented a data driven SOH estimation strategy based on Bayesian inference probabilistic model, where the efficacy of the scheme depends upon the data collected from the sensors. In [12], Shen *et.al.* proposed a model, which is designed for co-estimation of SOC, SOH, and SOF using EKF and RLS. In [13], wavelet analysis-based estimation strategy for obtaining the relative SOH between the battery cells is presented. The experimental data analysis done using neural network in [14], shows that there is nonlinear interdependence of minimum impedance level on the capacity loss for a battery, which is used to estimate the capacity of the cells.

A robust recurrent neural network-based data driven approach has been proposed in [15] for SOC and SOH estimations, where the knowledge of the battery parameters is not required for computation. In [16], advanced fault detection and isolation of battery units based on SOC and SOH has been proposed along with the design and development of a low cost BMS prototype. In [17], Xiong *et.al* presented a linear aging model developed based on parametric profile for forecasting the durability of an EV battery.

The review shows a robust BMS needs accurate and efficient algorithms to estimate the parameters like SOH-State of Health and SOC-State of Charge. A constant threshold for the SOC is used in the above works, while the health should depends on

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several parameters like environment, temperature, material etc. So this constant threshold cannot be used in all circumstances. Different applications can be derived from just estimating parameters like SOC and SOH.

The present paper addresses the problem of estimation of SOC and SOH parameters of the battery with the help of data collected of the batteries like voltage, current and temperature using sensors and also aim to develop an algorithm which would be having an adaptive threshold for SOC and can be used in all the circumstances.

The rest of the paper is organized as follows: Section II deals with the architecture of the proposed BMS. Section III presents the Estimation strategies for SOC and SOH. The adaptive algorithm for tuning the threshold for optimal SOC has been derived in Section IV. The result analysis and the conclusions are presented in Section V and Section VI respectively.

II. SYSTEM OVERVIEW

Over charging and discharging of electric vehicles batteries are one of the main reasons of reduction in battery lifespan. A robust battery management system, with efficient battery parameters estimation methods is most efficacious means by which the batteries can be safeguarded and can be used to its fullest potential. The objective of this work is to design a prototype of a battery management System (BMS) in Electric vehicles, which can monitor and protect the batteries of the vehicle by estimating parameters like SOH and SOC of the batteries with the help of system parametric data collected of the batteries such as voltage, current and temperature using sensors and also aim to develop an algorithm, which would be having an adaptive threshold for SOC and can be used in all the circumstances.

This system model can be broadly divided into four functions:

- 1) Data collection/ aquisition: The information about batteries like it's voltage, current and temperature is collected using different type of sensors
- 2) State Estimation: Estimation of parameters like SOC and SOH takes place with the help of efficient algorithms.
- 3) Determine the problem: After estimating the states, any irregularities are found out in any specific area and checking the estimated state with the adaptive threshold of SOH.
- 4) Action against the problem/ communicating the problem to the user:

Fig1 shows the complete block schematic of Battery Management System.

III. ESTIMATION OF SOC AND SOH

The state-of-charge (SOC) and the terminal resistance of the onboard battery can be estimated using UKF-Unscented Kalman Filter. The terminal resistance, R_0 , is a state of the estimator. The ratio of the released capacity $C_{releasable}$ of the battery to its rated capacity C_{rated} is termed as SOC: $SOC = C_{releasable} / C_{rated}$, which also indicates the peak amount of battery charge and the data regarding the rated

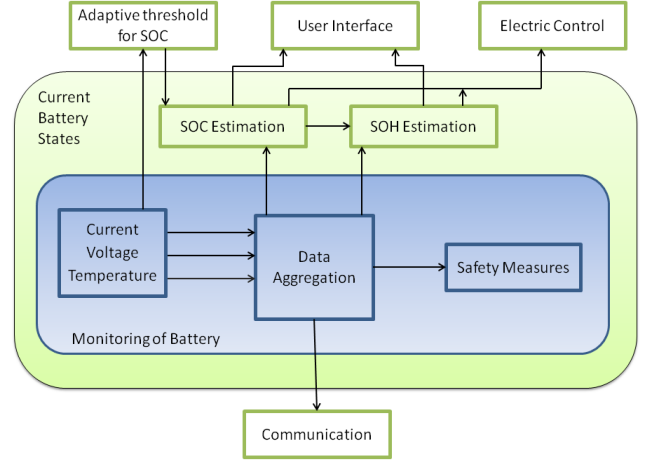


Fig. 1. Block schematic of Battery Management System

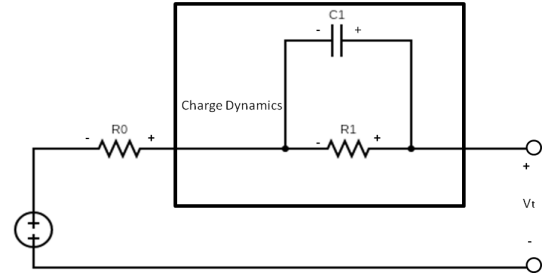


Fig. 2. Equivalent Circuit Diagram of a battery [6] [7]

capacity is usually provided by the manufacturers. The dynamics of the battery parameters derived based on the equivalent circuit given by Fig 2 are as follows [6] [7]:

$$\frac{dSOC}{dt} = \frac{-i}{3600AH(T)} \quad (1)$$

$$\frac{dV_1}{dt} = \frac{i}{C_1(SOC, T)} - \frac{V_1}{R_1(SOC, T)C_1(SOC, T)} \quad (2)$$

$$\frac{dR_0}{dt} = 0 \quad (3)$$

$$V_t = V_0(SOC, T) - iR_0 - V_1 \quad (4)$$

where i , T , AH , V_0 , V_t , R_1 , C_1 is the current, temperature, ampere-hour rating, no-load voltage, terminal voltage, first polarization resistance and RC capacitance connected in parallel respectively. Also τ_1 , is a time constant which is being used in the relation $C_1 = \tau_1 / R_1$.

For UKF algorithm, the following states for the non-linear functions are used [6] [7]:

$$s = [SOC \quad V_1 \quad R_0]^T \quad (5)$$

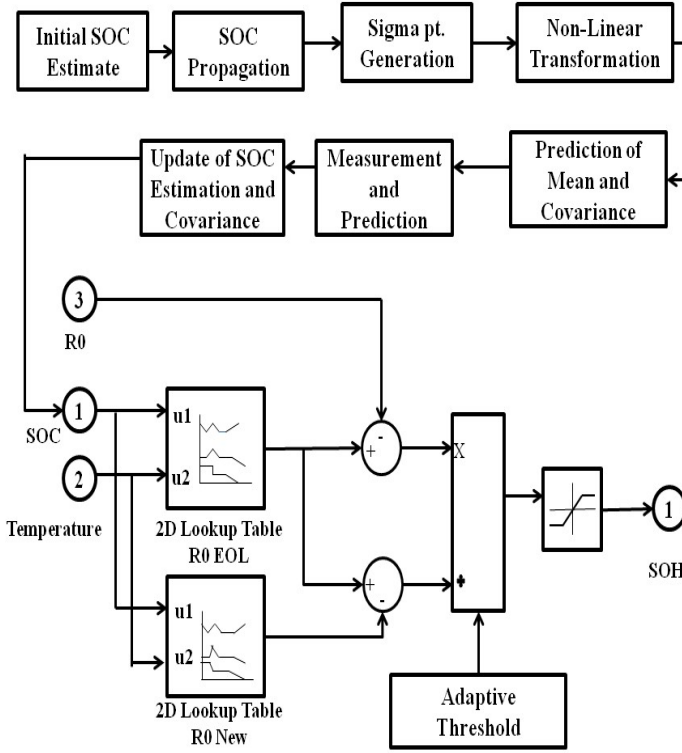


Fig. 3. Block schematic of the SOH and SOC estimation schemes

$$f(s, i) = \left[\frac{\frac{-i}{3600AH(T)}}{C_1(SOC, T) - \frac{V_1}{R_1(SOC, T)C_1(SOC, T)}} \right] \quad (6)$$

$$h(s, i) = V_0(SOC, T) - iR_0 - V_1 \quad (7)$$

The block schematic of the SOC and SOH estimation strategies for the robust battery management system is given in fig 3.

In an UKF algorithm, an approach called unscented transformation is used in which non-linear operations are applied on previously chosen sigma points. Then from the resultant points, mean and covariance matrix are formed whose accuracy is more. More detailed steps of the UKF algorithm are mentioned below [6] [7]:

- 1) Initializing the algorithm $s'(0|0)$ is the initial SOC estimate and $E'(0|0)$ be Initial covariance matrix. The mean weight and covariance weight are computed based on generated sigma points [6].

$$s^{(i)} = \begin{cases} s'(j+1|j), & i = 1 \\ s'(j+1|j) + (\sqrt{(d+\lambda)p(j|j)i}), & i = 2, 3, \dots, 2d+1 \\ s'(j+1|j) - (\sqrt{(d+\lambda)p(j|j)i}), & i = d+2, 3, \dots, 2n+1 \end{cases}$$

$$W_m^{(i)} = \begin{cases} \frac{\lambda}{\lambda+d} \\ \frac{\lambda}{\lambda+d} + (1 - \alpha^2 + \beta) \end{cases} \quad (8)$$

where d is the dimension of the estimated vector, $\lambda = \alpha^2(d + \kappa) - d$, $\alpha \in [0, 1]$ is the distance between the the sigma points and it's mean. $\beta = 2$ and $\kappa = 0$, for a normal distribution.

- 2) Estimation of the state matrix [6] [7]

$$\hat{s}(i)(j+1|j) = f(\hat{s}(i)(j|j), i(j))\hat{s}(j+1|j) = \sum_{i=1}^{2d+1} W_m^{(i)} \mathbf{W}(i) m\hat{s}(i)(j+1|j) \quad (9)$$

- 3) Estimate the measured variables: $V_t^{(i)}(j+1|j)$
- 4) Estimate the measurement covariance (E_y) as well as covariance between the state and measurement (E_{sy}):
- 5) Compute the gain of Kalman filter

$$j(j+1) = E_{sy}E_y^{-1} \quad (10)$$

- 6) Updation of covariance and state matrix [6] [7]:

$$\hat{\mathbf{s}}(j+1|j+1) = \hat{\mathbf{s}}(j+1|j) + \mathbf{j}(j+1)(V_t(j+1) - \hat{V}_t(j+1|j)) \quad (11)$$

$$\mathbf{E}(j+1|j+1) = \mathbf{E}(j+1|j) - \mathbf{K}(j+1)\mathbf{E}_y\mathbf{K}^T(j+1)$$

SOH is the state estimated using the previous step. SOH indicates the relative state of battery with respect to the ideal conditions, and provides knowledge about the battery's life. The battery aging can be interpreted from its terminal resistance (R_0). We can estimate the terminal resistance by using the SOC Estimator.

$$SOH = \frac{R_{EOL} - R_0}{R_{EOL} - R_{0,new}} \quad (12)$$

where R_{EOL} is the end-of-life resistance and $R_{0,new}$ is the terminal resistance, when the battery is new.

IV. ADAPTIVE THRESHOLD OF SOC FOR OPTIMAL BATTERY

An novel adaptive algorithm is proposed for tuning the threshold on the SOC to ensure optimal performance of the battery. From [2], $SOH = \frac{C}{C_n}$, where C and C_n represent the actual and nominal capacities of the battery respectively. $\frac{C}{C_n} \leq 1$ as $C_n \geq C$. $\frac{C}{C_n}$ should always be maximum for an optimal battery. In other terms we can also write that, $(1 - \frac{C}{C_n})$ should be minimum for an optimal battery. Therefore, our objective is to minimize $(1 - \frac{C}{C_n})$.

Consider a Lyapunov Function:

$$V = \frac{1}{2} \left(1 - \frac{C}{C_n}\right)^2 \quad (13)$$

Differentiating the above equation:

$$V' = \left(1 - \frac{C}{C_n}\right) \left(\frac{-1}{C_n}\right) C' \quad (14)$$

Also,

$$C_{\alpha,\beta} = \frac{\Delta Ah}{\Delta SOC} = \frac{\int_{t_\beta}^{t_\alpha} I_{cell}(t) dt}{SOC(t_\alpha) - SOC(t_\beta)} \quad (15)$$

Here, η is the coulomb efficiency, which is a constant. Putting these values in equation(14):

$$V' = (1 - \frac{C}{C_n}) \left\{ \frac{-\Delta SOC \cdot I_{cell}(t)}{C_n} - \frac{1}{C_n} \int_{t_\beta}^{t_\alpha} I_{cell}(t) dt \eta \left[\frac{-I_\alpha}{Q_{st}} + \frac{-I_\beta}{Q_{st}} \right] \right\} \quad (16)$$

$(1 - \frac{C}{C_n})$ will always be positive. Hence, for the V' to be stable and negative semi-definite, according to Lyapunov theory.

$$\left\{ \frac{-\Delta SOC \cdot I_{cell}(t)}{C_n} - \frac{1}{C_n} \int_{t_\beta}^{t_\alpha} I_{cell}(t) dt \eta \left[\frac{-I_\alpha}{Q_{st}} + \frac{-I_\beta}{Q_{st}} \right] \right\} \leq 0 \quad (17)$$

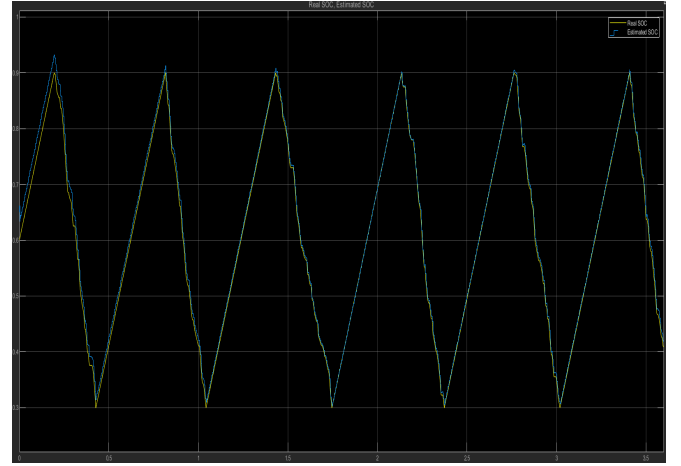
$$\Delta SOC^* \geq \left[\frac{C_n}{I_{cell}(t)} \right] \left[\int_{t_\beta}^{t_\alpha} I_{cell}(t) dt \frac{(-I_\alpha + I_\beta)}{Q_{st}} \right] \quad (18)$$

ΔSOC^* is the change in State of Charge for an optimal battery.

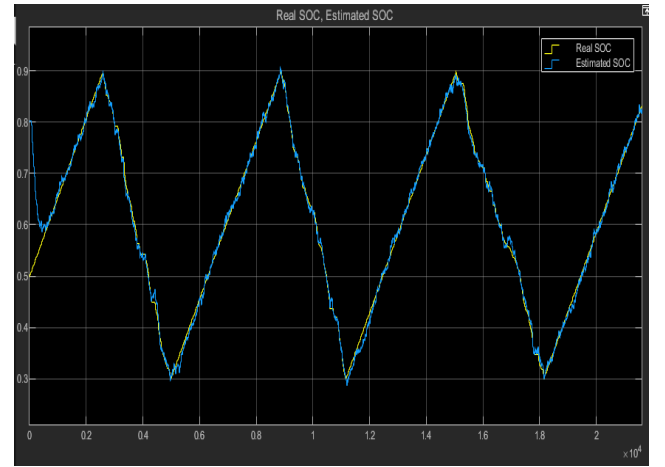
V. RESULTS AND DISCUSSION

The state of charge of a battery was estimated for diverse conditions of the battery and other parameters such as initial voltage, self discharging resistance of the battery, terminal operating voltage, terminal resistance, open circuit voltage, number of discharge cycles. For SOC estimation the standard UKF filter was used. Fig4 (a), fig4 (b) and fig4 (c) shows the plot of real and estimated SOC at different initial conditions and different battery parameters. Here the curve with blue line denotes the estimated SOC while yellow colour denotes the actual SOC. In fig4 (a), the initial voltage was taken as 0.6V and the self charge resistance was taken as 6000 ohms and number of discharge cycles was taken 10. In fig4 (b), the initial voltage was taken as 0.5V, the self charge resistance was taken as 7000 ohms and number of discharge cycles were 10. In fig4 (c), the initial voltage is 0.5V but the self discharge resistance was 8000 ohms and the number of self discharge cycles were 15. The simulations are performed using Matlab.

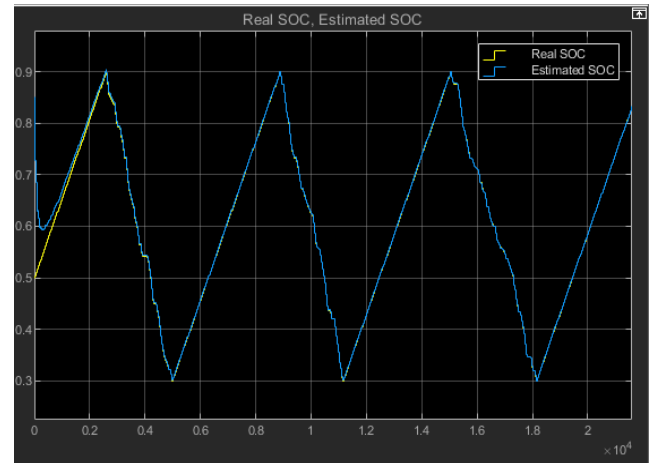
The SOC estimated from the earlier block is to estimate the SOH of battery and plotted the graph for the same. The first plot in the figure fig5 shows the SOH estimated with respect to time and the second plot in fig5 shows the estimated Terminal Resistance(R_0) with the respect to time. Fig 7 and fig8 show the estimated SOH at different initial conditions and battery parameters, that were used in the plotting of SOC. It can be seen from all the SOH plots that the initial terminal resistance of the battery to be 0 or close to 0. It just indicates the optimal condition of the battery, when they are used for the first time while there efficiency decreases as they are used further. As from the graphs, it can be easily observed that just shows that with time as the terminal resistance among the battery



(a)



(b)



(c)

Fig. 4. Estimated SOC trajectories for various conditions

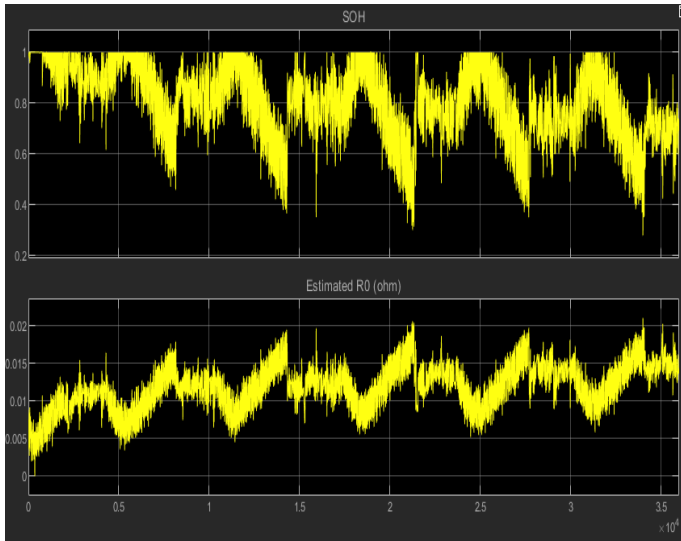


Fig. 5. Plot-1 of estimated SOH

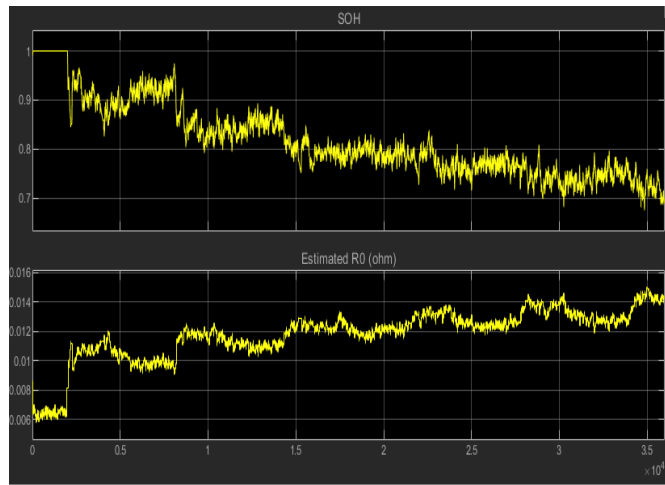


Fig. 6. Plot-2 of estimated SOH

increases the state of health (SOH) decreases relatively which leads to adverse affects.

The threshold value of SOC for an optimal battery that is developed in this paper ensures that if the change in SOC for two given time instances is lesser than the following term than the given battery in optimal state and is good to use. The equation developed for the threshold is based on Lyapunov Theory of Stability.

VI. CONCLUSION

A prototype system is developed, which not only increases the efficiency of batteries used in electric vehicles but also reduces the chances of accidents caused due to battery failure. Strategies and optimization techniques to improve the performance of batteries is the need of the hour, and which is being covered by working on parameters like SOC and SOH of the battery. By using efficient ways for estimating the values of SOC and SOH, which will lead to efficient usage of battery

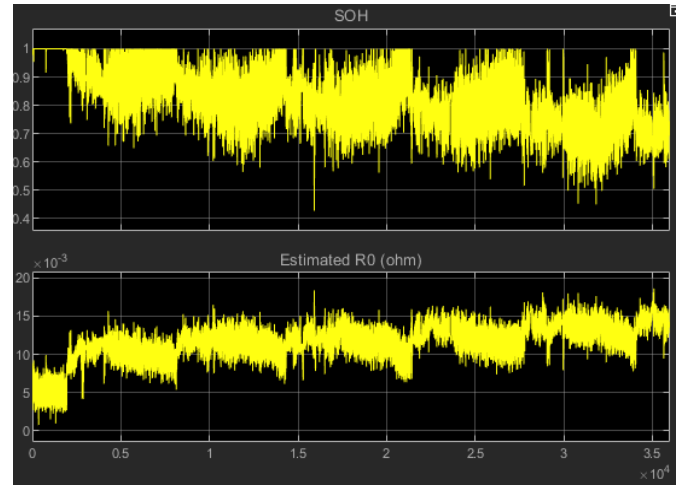


Fig. 7. Plot-3 of estimated SOH

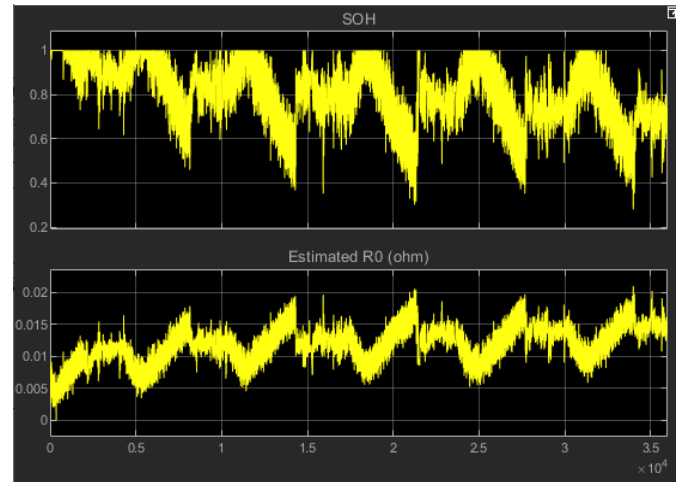


Fig. 8. Plot-4 of estimated SOH

and enabling them for a longer run and moreover, it reduces the risk factor related to batteries in an electric vehicle.

Also the proposed adaptive threshold for the SOC for the battery is suited for diverse operating conditions, which ensures that the given battery is in optimal state or not. In other words, it gives the information about the end-of-life of the battery.

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